CEVE 421/521 Schedule

> **ℹ Note**
>
> Check the page for each reading to see discussion questions. You will often be encouraged to focus only on part of relevant papers.
>
> **Final Project:** Climate Plan Audit *(in prep)* — Deliverables are due throughout the semester (see schedule below).
>
> **CEVE 521 students:** You will lead one Paper Discussion session. See Graduate Seminar Requirements.

Week	Date	What
1	Mon., Jan. 12	**Lecture:** Intro to Climate Risk Management slides
	Wed., Jan. 14	**Topical Discussion:** T. Frank [1] and S. Crawford [2]
	Fri., Jan. 16	**Lab 1:** Julia Setup & Hello World
2	Mon., Jan. 19	*Martin Luther King, Jr. Day (No Class)*
	Wed., Jan. 21	**Lecture:** Hazard, Exposure, Vulnerability slides
	Fri., Jan. 23	**Lab 2:** Hazard-Damage Convolution
3	Mon., Jan. 26	**Lecture:** Climate Variability, Extremes, and Change slides
	Wed., Jan. 28	**Paper Discussion:** Climate Extremes (IPCC)
	Fri., Jan. 30	**Lab 3:** Sea-Level Rise Scenarios and Uncertainty / **Project:** Topic Proposal Due
4	Mon., Feb. 2	**Lecture:** Uncertainty, Monte Carlo & Risk Metrics slides
	Wed., Feb. 4	**Paper Discussion:** ICOW Framework slides
	Fri., Feb. 6	**Lab 4:** Monte Carlo & Risk Metrics
5	Mon., Feb. 9	**Lecture:** Valuation & Benefit-Cost Analysis slides
	Wed., Feb. 11	**Paper Discussion:** Discounting & BCA
	Fri., Feb. 13	*No Class*

Week	Date	What
6	Mon., Feb. 16	**Lecture:** Optimal Static Decisions slides
	Wed., Feb. 18	**Exam 1** (Covers Weeks 1-4)
	Fri., Feb. 20	**Lab 5:** Simulation-Optimization
7	Mon., Feb. 23	**Lecture:** Value of Information
	Wed., Feb. 25	**Lecture:** Value of Imperfect Information / **Lecture:** Global Sensitivity Analysis slides
	Fri., Feb. 27	**Lab 6:** Value of Partial Information
8	Mon., Mar. 2	**Lecture:** Robustness & Deep Uncertainty slides / **Project:** Teams & Topics Assigned
	Wed., Mar. 4	**Paper Discussion:** Robustness & Deep Uncertainty
	Fri., Mar. 6	**Lab 7:** Robustness Metrics
9	Mon., Mar. 9	**Workshop:** Final Project Work Session
	Wed., Mar. 11	**Lecture:** Trade-offs & Values slides
	Fri., Mar. 13	**Exam 2** (Covers Weeks 5-8)
	Mar. 16-20	*Spring Break*
10	Mon., Mar. 23	**Lecture:** Exploratory Modeling & Scenario Discovery slides
	Wed., Mar. 25	**Paper Discussion:** Exploratory Modeling & Scenario Discovery
	Fri., Mar. 27	**Lab 8:** Scenario Discovery / **Project:** Memo 1 Due
11	Mon., Mar. 30	**Lecture:** Sequential Decision Making slides
	Wed., Apr. 1	**Paper Discussion:** Sequential Decision Making
	Fri., Apr. 3	**Lab 9:** Real Options & the Parking Garage
12	Mon., Apr. 6	**Lecture:** Multi-Objective Optimization slides
	Wed., Apr. 8	**Paper Discussion:** Multi-Objective Optimization
	Fri., Apr. 10	**Lab 10:** Pareto Fronts *(in prep)* / **Project:** Memo 2 Due
13	Mon., Apr. 13	**Lecture:** Case Studies slides *(in prep)*

Week	Date	What
	Wed., Apr. 15	**Paper Discussion:** Case Studies *(in prep)*
	Fri., Apr. 17	**Project:** Slides Due *(in prep)*
14	Mon., Apr. 20	Final Project Presentations *(in prep)* (Teams 1 & 2)
	Wed., Apr. 22	Final Project Presentations *(in prep)* (Teams 3 & 4)
	Fri., Apr. 24	*Backup/overflow day*
Finals	TBD	**Exam 3** (Covers Weeks 9-13) / **Project:** Written Report Due *(in prep)*

Bibliography

[1] T. Frank, "Bold New Jersey Shore Flood Rules Could Be Blueprint for Entire U.S. Coast," *Scientific American*, Aug. 2022, Accessed: Jan. 11, 2023. [Online]. Available: https://www.scientificamerican.com/article/bold-new-jersey-shore-flood-rules-could-be-blueprint-for-entire-u-s-coast/

[2] S. Crawford, "Floods and Storms Are Ravaging the Jersey Shore. Why Do We Keep Building It Back?." Accessed: Dec. 05, 2025. [Online]. Available: https://www.rollingstone.com/culture/culture-features/floods-storms-jersey-shore-beach-rebuilding-1235477927/